

12.164

EE25BTECH11047 - RAVULA SHASHANK REDDY

October 13, 2025

Question:

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Find the maximum number of linearly independent eigenvectors of $\mathbf{A}$.

Solution:

Since $\mathbf{A}$ is a triangular matrix

$$\lambda = 1, 1, 1 \tag{1}$$

$$(\mathbf{A} - \mathbf{I})\mathbf{x} = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{x} = 0, \quad \mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \tag{2}$$

$$\Rightarrow \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{3}$$

$$R_1 \rightarrow R_1 - R_2 \Rightarrow \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{4}$$

$$z = 0, y = 0 \tag{5}$$

$$\mathbf{x} = \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} = x \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \tag{6}$$

Maximum number of linearly independent eigenvectors = 1